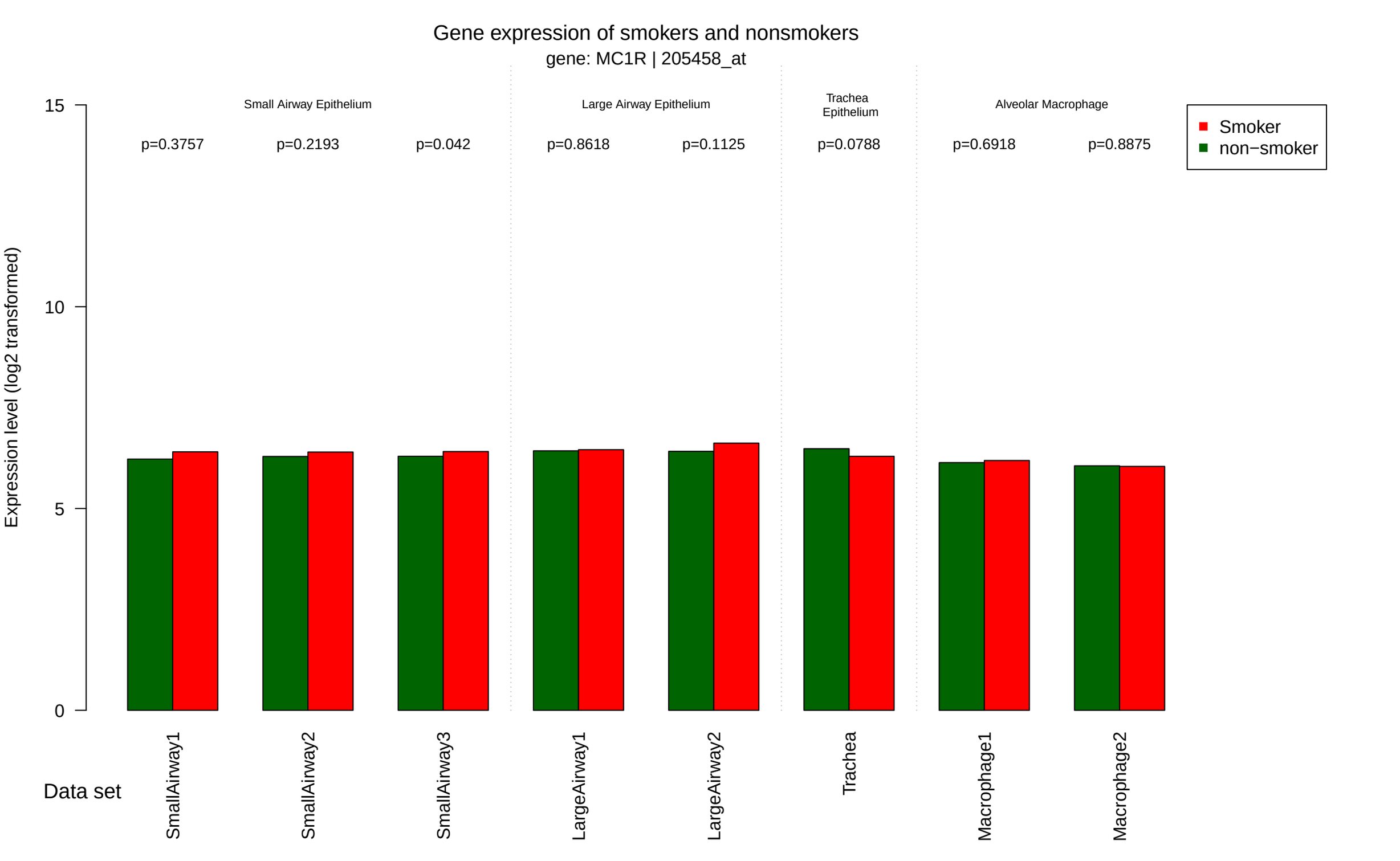
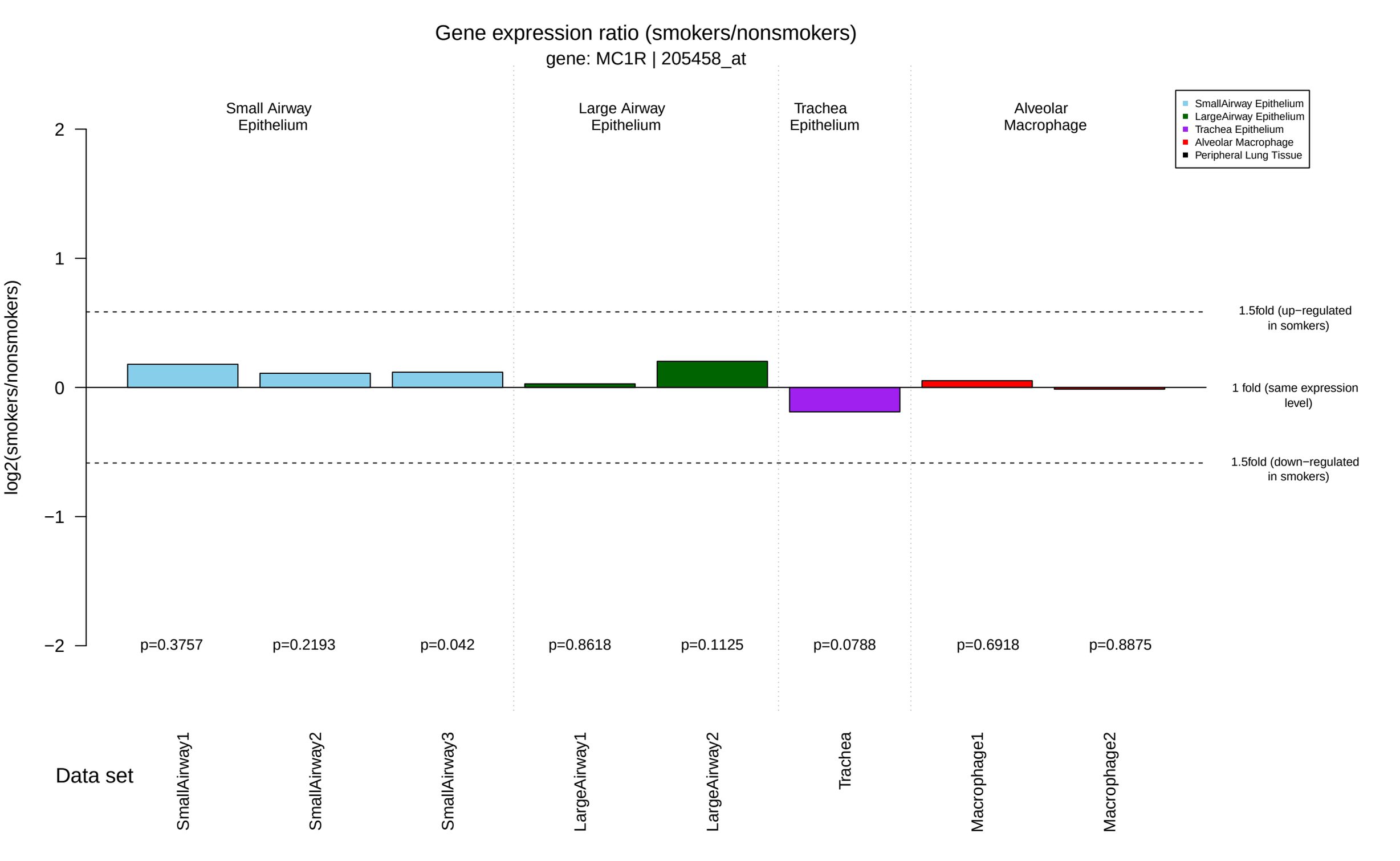
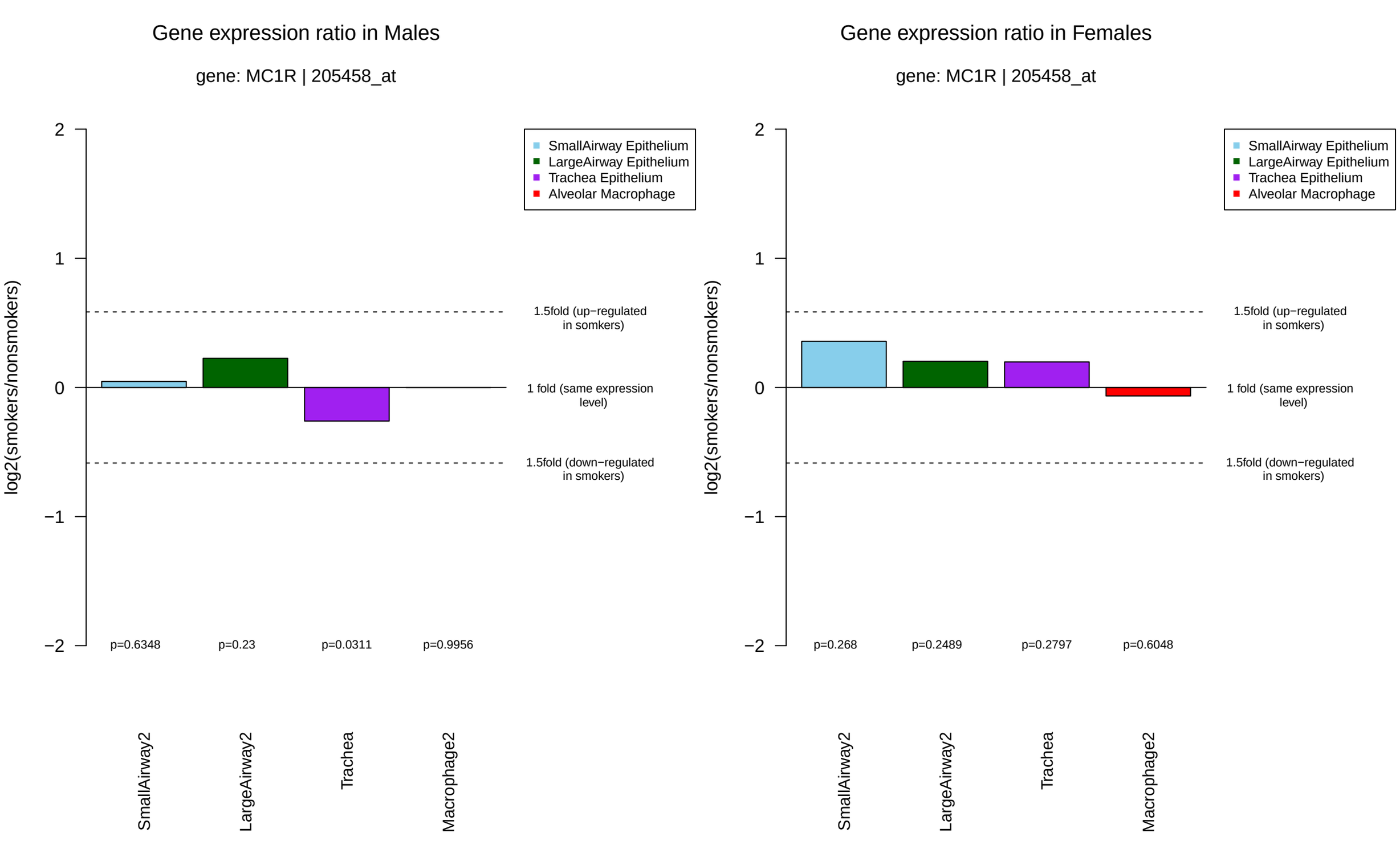
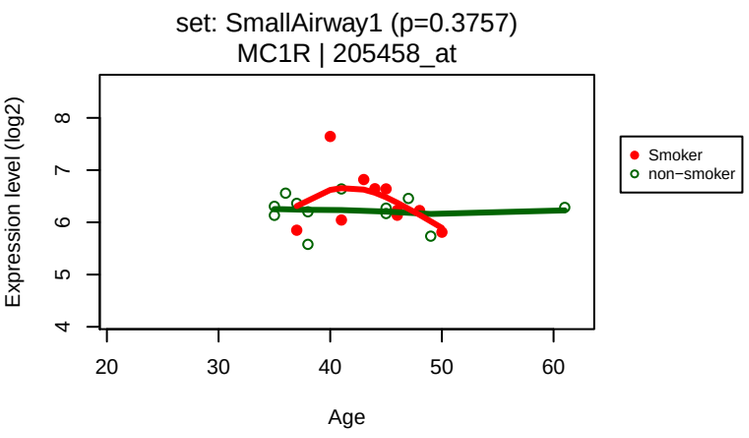
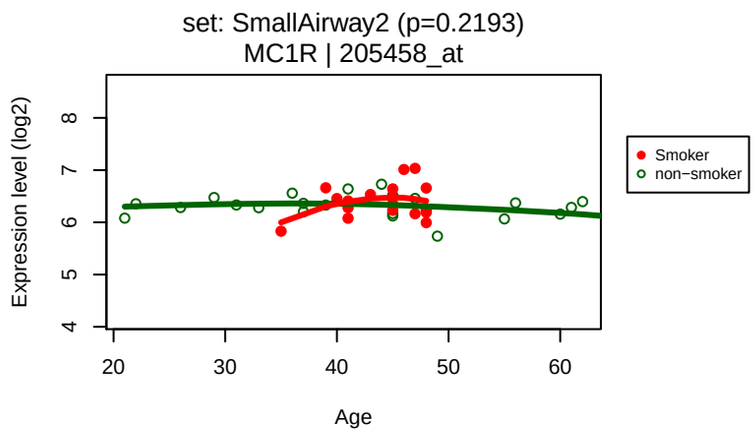
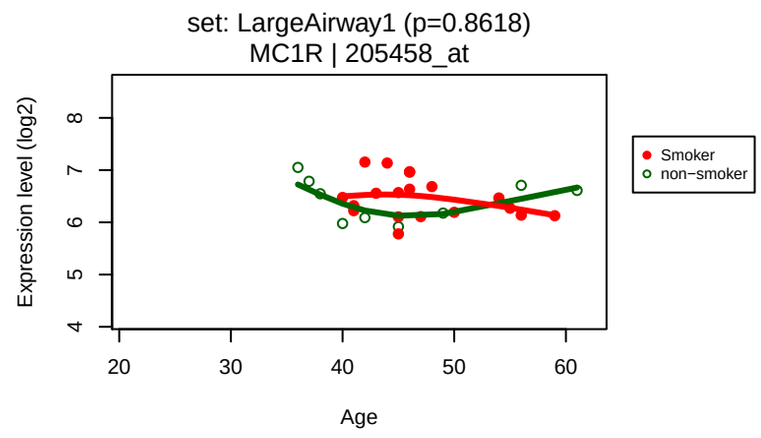

Gene expression of smokers and nonsmokers
gene: MC1R | 205458_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.3757
p=0.2193
p=0.042
p=0.8618
p=0.1125
p=0.0788
p=0.6918
p=0.8875
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

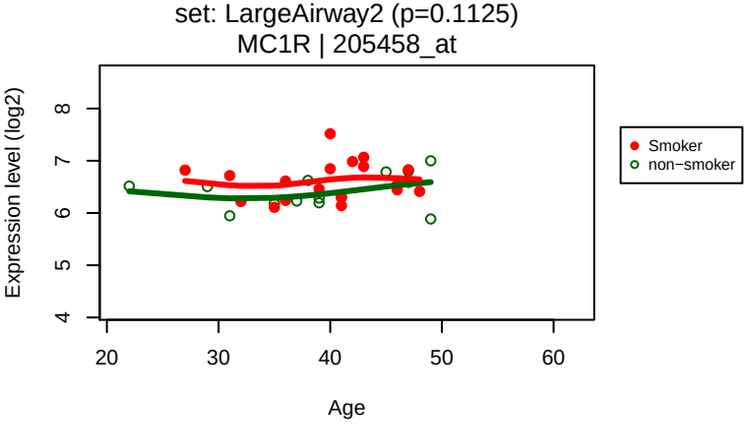

Gene expression ratio (smokers/nonsmokers)
gene: MC1R | 205458_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.3757
p=0.2193
p=0.042
p=0.8618
p=0.1125
p=0.0788
p=0.6918
p=0.8875
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

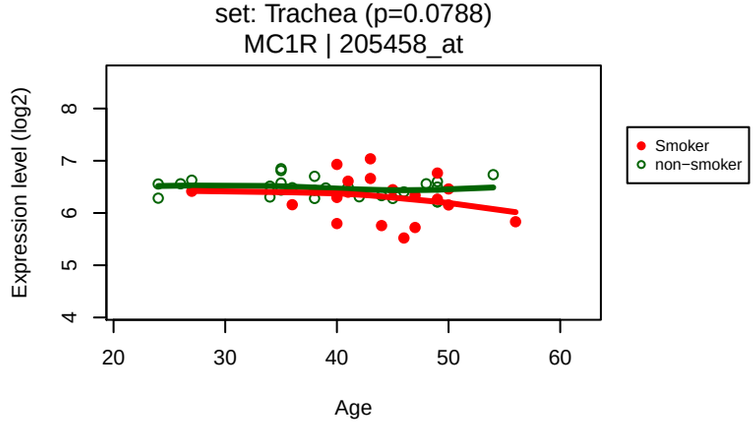

Gene expression ratio in Males
gene: MC1R | 205458_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.6348
p=0.23
p=0.0311
p=0.9956
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: MC1R | 205458_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.268
p=0.2489
p=0.2797
p=0.6048
SmallAirway2
LargeAirway2
Trachea
Macrophage2

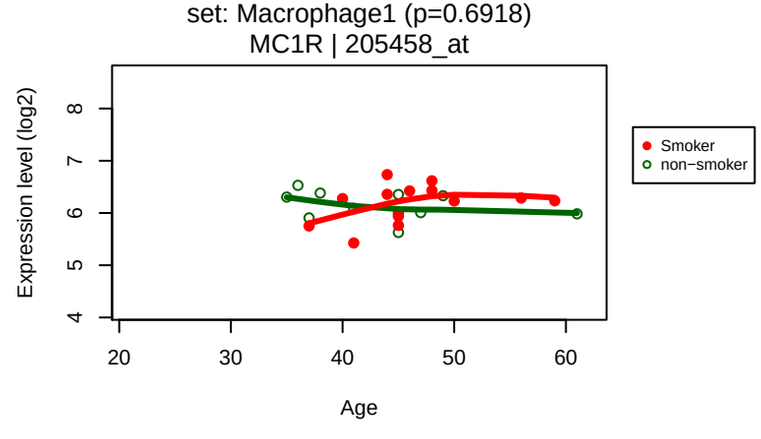

set: SmallAirway1 (p=0.3757)
MC1R | 205458_at
Expression level (log2)
Age
Smoker
non−smoker

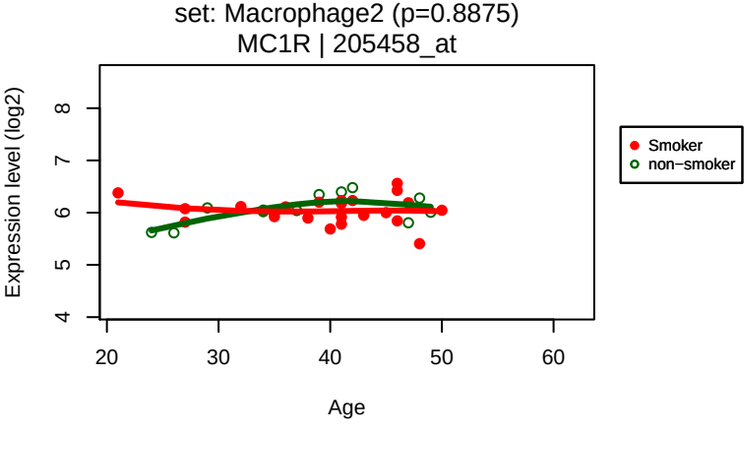

set: SmallAirway2 (p=0.2193)
MC1R | 205458_at
Expression level (log2)
Age
Smoker
non−smoker

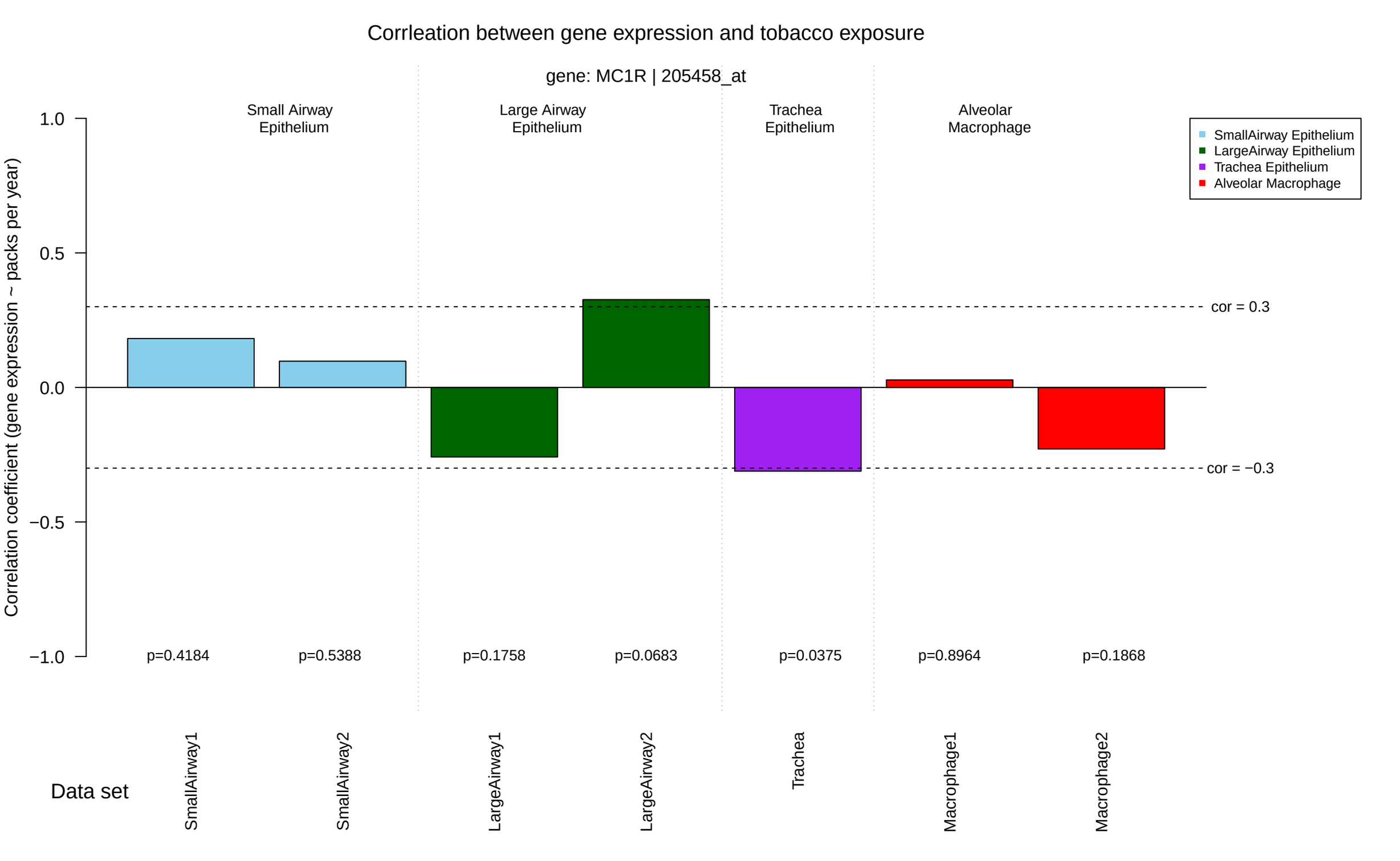

set: LargeAirway1 (p=0.8618)
MC1R | 205458_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.1125)
MC1R | 205458_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.0788)
MC1R | 205458_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.6918)
MC1R | 205458_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.8875)
MC1R | 205458_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: MC1R | 205458_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.4184
p=0.5388
p=0.1758
p=0.0683
p=0.0375
p=0.8964
p=0.1868
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2